

C O N F I D E N T I A L

URINALYSIS AI

Computer Vision Pregnancy Detection System

Clinical & Technical Overview

For Medical & Scientific Review

Version 1.0 | May 2026

101 Subjects | 1,222 Images | 68 Video Clips

RESEARCH USE ONLY

Not a diagnostic device. Experimental probability model.

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1. Executive Summary

Urinalysis AI is an experimental computer vision system that detects pregnancy from smartphone photographs and video clips of urine samples. Using machine learning algorithms trained on 101 subjects (46 confirmed pregnant, 55 confirmed non-pregnant), the system achieves an Area Under the Receiver Operating Characteristic Curve (AUC-ROC) of 0.774 in rigorous 5-fold cross-validation with subject-level grouping to prevent data leakage.

The system analyzes 179 quantitative features extracted from each urine sample, spanning color analysis (RGB, CIELAB, hue, saturation), optical properties (turbidity, transparency, concentration darkness), surface dynamics (foam, bubbles, texture), and temporal behavior captured through video (color drift, brightness trends, optical flow coherence, foam dissipation patterns).

The core hypothesis is that pregnancy-related hormonal changes (primarily hCG, progesterone, and estrogen metabolites) alter the physical and optical properties of urine in ways that are detectable by computational imaging, even when not visible to the human eye. This represents a novel approach to non-invasive, zero-cost pregnancy screening using only a consumer smartphone camera.

AUC-ROC	Sensitivity	Specificity	Subjects
0.774	65.2%	78.2%	101 (46P / 55N)

2. Clinical Context & Rationale

2.1 The Problem

Conventional pregnancy tests detect human chorionic gonadotropin (hCG) in urine via lateral flow immunoassay. While highly accurate (sensitivity >99%, specificity >99%), they require: (a) purchase of a physical test kit, (b) a private setting to perform the test, and (c) correct interpretation of results. In developing regions, cost, availability, and stigma create barriers to early pregnancy detection.

Delayed pregnancy detection is associated with adverse maternal and fetal outcomes: missed prenatal care windows, continued exposure to teratogenic substances, delayed management of ectopic pregnancy, and reduced time for reproductive decision-making.

2.2 The Hypothesis

Pregnancy induces systemic metabolic, hormonal, and renal changes that alter urine composition. These changes include: elevated hCG and its metabolites, increased progesterone and estrogen, altered glomerular filtration rate (GFR), changes in urine osmolality and specific gravity, modified excretion of amino acids, vitamins, and minerals, and pH shifts due to metabolic alkalosis of pregnancy.

We hypothesize that these biochemical changes produce measurable differences in the optical, colorimetric, and physical properties of urine that can be captured by smartphone camera sensors and classified by machine learning models. This is analogous to how trained urologists can sometimes infer clinical information from visual urine inspection, but with computational precision exceeding human perceptual limits.

2.3 Prior Art & Novelty

Urine color analysis has been used clinically for centuries (uroscopy), and modern dipstick analysis quantifies color changes for pH, protein, glucose, and other analytes. Recent work in computational urinalysis has focused on digitizing dipstick readings. Our approach is fundamentally different: we analyze the urine itself, not a reagent reaction, using raw optical properties and temporal dynamics captured by video.

To our knowledge, this is the first system attempting pregnancy detection through direct computational analysis of urine appearance without chemical reagents.

3. Methodology

3.1 Study Design

Cross-sectional observational study. Participants provided urine samples in clear plastic cups, photographed with their personal smartphones under varied lighting conditions (indoor LED, natural daylight, mixed). Each sample was documented with 2-4 photographs (top view, side view) and one video clip (5-15 seconds). A paper questionnaire captured clinical metadata.

3.2 Participants

Metric	Positive (Pregnant)	Negative (Control)
N	46	55
Age range	18 - 37	11 - 40
Gestational age	2 - 36 weeks	N/A
First morning urine	14%	19%
Pregnancy confirmation	Blood test / Ultrasound / Home test	Self-reported non-pregnant
Photos	447 total	
Video clips	68 (36P + 32N)	

3.3 Feature Extraction Pipeline

Each image undergoes automated processing to extract 32 quantitative features per frame:

- **Colorimetric features:** Mean R, G, B channels; brightness; dominant hue; hue spread; saturation; yellowness index; green-to-yellow ratio (GY ratio)
- **CIELAB color space:** L* (lightness), a* (green-red axis), b* (blue-yellow axis), chroma (color intensity). CIELAB is perceptually uniform and models human color vision more accurately than RGB.
- **Optical properties:** Turbidity (coefficient of variation of brightness), texture score (Laplacian variance), edge intensity, contrast, color variance, concentration darkness proxy (inverse brightness weighted by saturation)
- **Spectral ratios (Layla features):** Custom spectral ratios (R/(G+B), R/G, R/B) with statistical moments (mean, std, skewness, dominant) designed to capture subtle spectral shifts in urine that may correlate with hormonal metabolites.
- **Calibration metrics:** Color card detection, Delta-E (color accuracy), estimated correlated color temperature (CCT) for lighting normalization.

For video clips, 42 additional temporal features are computed: color channel trends over time, brightness stability, hue drift, optical flow coherence (surface dynamics), texture variation, and foam/bubble dissipation patterns. These temporal features capture how the urine sample behaves dynamically, which may reflect pregnancy-related changes in surface tension, protein content, and dissolved solutes.

Per-subject aggregation computes mean, standard deviation, minimum, and maximum across all images, expanding 32 raw features to 137 photo features. Combined with 42 video features: 179 total features per subject.

3.4 Machine Learning Architecture

The production model (V5) uses a three-stage pipeline:

- **Feature Selection:** SelectKBest with ANOVA F-statistic (f_{classif}), retaining $k=50$ most discriminating features from the 179-dimensional feature space.
- **Normalization:** StandardScaler (zero mean, unit variance) to ensure equal feature contribution regardless of scale.
- **Classification:** Soft Voting Ensemble of three diverse classifiers: Logistic Regression ($C=0.01$, linear decision boundary), Extra-Trees Classifier (500 trees, max depth 5, non-linear ensemble), and Support Vector Machine (RBF kernel, $C=1.0$, non-linear margin maximization). Soft voting averages predicted probabilities across all three classifiers.

Confounder control features (age, first morning urine, recent coffee intake, fluid consumption, health conditions) are included to help the model account for non-pregnancy factors that affect urine appearance. Pregnancy-indicative features (gestational age, folic acid use) are explicitly excluded to prevent label leakage and ensure the model learns from urine appearance, not metadata.

3.5 Evaluation Protocol

5-fold Group K-Fold cross-validation with subject-level grouping. This ensures that all images from a single subject appear in either the training or test set, never both, preventing data leakage from intra-subject similarity. Each fold trains on approximately 80 subjects and tests on approximately 20.

Primary metric: Area Under the ROC Curve (AUC-ROC), which measures discrimination ability across all possible classification thresholds. Secondary metrics: sensitivity (true positive rate), specificity (true negative rate), and Youden's J statistic (sensitivity + specificity - 1) at the optimal threshold.

4. Results

4.1 Model Performance

Model Configuration	AUC	Sensitivity	Specificity	Youden J
Photo features only	0.734	63.0%	72.7%	0.357
Photo + Video	0.739	63.0%	76.4%	0.394
Photo + Video + Confounders	0.774	65.2%	78.2%	0.434

The addition of confounder control features (age, hydration status, coffee intake, health conditions) improved AUC from 0.739 to 0.774, confirming that these non-pregnancy factors add noise to the urine signal that the model can partially correct for.

4.2 Top Discriminating Features

The 10 features with highest ANOVA F-scores reveal what the system detects:

#	Feature	F-score	Type	Clinical Interpretation
1	g_mean_std	21.0	Photo	Green channel variability across images
2	lab_L_std	19.6	Photo	Lightness consistency (concentration stability)
3	gy_ratio_mean	15.8	Photo	Green-to-yellow spectral balance
4	conc_darkness_std	15.5	Photo	Concentration darkness variability
5	brightness_std	15.5	Photo	Brightness consistency
6	t_r_ratio_complexity	15.0	Video	Red channel temporal complexity
7	t_hue_mean	14.5	Video	Average hue over time (color drift)
8	r_mean_std	13.7	Photo	Red channel variability
9	b_mean_std	13.5	Photo	Blue channel variability
10	brightness_cv	13.3	Photo	Coefficient of variation of brightness

A striking finding: 7 of the top 10 features measure variability (standard deviation, coefficient of variation) rather than absolute values. This suggests that the consistency of urine optical properties across multiple photos is more diagnostically informative than any single measurement. Pregnant urine may exhibit different optical stability patterns compared to non-pregnant urine, possibly related to protein content, hormone metabolite concentrations, or surface tension differences affecting light interaction.

4.3 Confounder Analysis

The following factors were identified as significant confounders affecting urine appearance independently of pregnancy status:

Confounder	Effect on Urine	Prevalence	Impact
Hydration status	Dilution / concentration	Variable	STRONG
Time of day	Morning = concentrated	85% non-morning	MODERATE
Coffee intake	Mild diuretic, color shift	50% of negatives	MODERATE
Iron supplements	Darker coloration	6% of positives	MODERATE
Vitamin B	Yellow intensification	12% overall	MILD
Fever	Concentrated, darker	5% of negatives	STRONG
UTI	Turbidity, color change	5% of negatives	STRONG

5. System Capabilities

5.1 What the System Can Detect Today

- Pregnancy probability score (0.0 to 1.0) from a single smartphone photo or video of a urine sample in a clear cup
- Urine concentration estimation (proxy for hydration status) via saturation-weighted inverse brightness
- Spectral shift detection in the red/green/blue channels that may correlate with hormonal metabolite presence
- Temporal behavior analysis from video: foam dissipation rate, surface stability, color drift over time
- Confounder-aware prediction that accounts for hydration, caffeine, age, and health conditions
- Sample quality assessment: lighting condition estimation, cup type detection, image suitability scoring

5.2 Medical Terminology & Measurements

The system operates with clinical-grade colorimetric precision using CIELAB ($L^*a^*b^*$) color space, which is the international standard for color measurement in medical diagnostics (ISO 11664). It computes Delta-E color differences, correlated color temperature (CCT) for illuminant characterization, and spectral ratios calibrated against reference color cards when available.

Feature naming follows established conventions: turbidity (nephelometric equivalent via pixel variance), concentration proxy (aligned with urine specific gravity estimation), and chromatic indices (comparable to urine color chart scoring systems such as the Armstrong scale).

5.3 Analytical Framework

The system represents a convergence of four scientific domains:

- **Computational urinalysis:** Quantitative analysis of urine optical properties exceeding human visual discrimination (>16 million color values vs. ~10 categories on clinical color charts)
- **Temporal microanalysis:** Video-based tracking of dynamic urine properties (foam, surface tension, settling behavior) at 30 frames/second
- **Ensemble machine learning:** Three complementary classifiers (linear, tree-based, kernel-based) combined via probability averaging for robust prediction
- **Confounder-aware modeling:** Explicit integration of known confounding variables to isolate the pregnancy-specific signal from hydration, dietary, and pathological noise

6. Vision & Development Roadmap

6.1 The Vision

Every woman with a smartphone should be able to perform an instant, free, private pregnancy screening by photographing a urine sample. No test kit purchase. No pharmacy visit. No waiting. No stigma. Just open the app, take a photo, and receive a probability score within seconds.

This does not replace clinical confirmation, it precedes it. A high-probability result triggers a recommendation to seek medical confirmation. A low-probability result provides reassurance. The system serves as a digital triage layer that democratizes access to pregnancy awareness.

6.2 Development Phases

Phase	Target	Requirements	Status
1. Proof of Concept	AUC > 0.70, N > 50	Basic pipeline, initial data	COMPLETE (AUC=0.774, N=101)
2. Clinical Pilot	AUC > 0.85, N > 500	Controlled collection, standardized protocol	NEXT
3. Validation Study	AUC > 0.90, N > 2,000	Multi-site, IRB approval, diverse population	Planned
4. Regulatory Path	FDA / CE clearance	Clinical trial, regulatory submission	Future

6.3 Immediate Next Steps (Phase 2)

- **Standardized collection protocol:** First morning urine only, standardized clear cup, controlled lighting via a simple photo box (cardboard + LED strip), 30-second video minimum
- **Quantitative ground truth:** Serum Beta-hCG levels (mIU/mL) for positive subjects, enabling regression modeling and trimester-specific calibration
- **Scale to 500+ subjects:** Including first trimester (<8 weeks), diverse age groups, and subjects with common confounders (fever, UTI, supplements)
- **Deep learning exploration:** Convolutional neural networks (CNN) operating on raw image pixels rather than hand-crafted features, potentially capturing patterns invisible to feature engineering
- **Clinical advisor engagement:** Ob/Gyn and clinical pathology consultation for protocol design, outcome validation, and regulatory guidance

7. For the Reviewing Physician

We are seeking clinical guidance on the following:

- **Biological plausibility:** Are the optical changes we detect consistent with known pregnancy-related urinary changes? Which metabolites or hormonal byproducts would most plausibly alter urine color, turbidity, or surface properties?
- **Collection protocol optimization:** What clinical standards should we adopt for urine collection timing, volume, and handling to maximize signal and minimize pre-analytical variation?
- **Ground truth validation:** Should we pair each sample with quantitative serum Beta-hCG, urine dipstick analysis, or both? What is the minimum clinical confirmation standard?
- **Confounder identification:** Beyond what we have identified (hydration, supplements, UTI, fever), what other conditions could mimic or mask pregnancy-related urine changes?
- **Clinical utility assessment:** At what performance level (AUC, sensitivity, specificity) would this system have clinical value as a screening tool? What is the acceptable false negative rate for a pre-screening application?
- **Regulatory pathway:** Would this classify as a medical device, wellness product, or screening tool under current FDA/CE frameworks? What evidence standard would be required?

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